Homework 3
Due: Thursday, 20th June 2024 at 1:00 PM

Instructions. Provide your answers **and an explanation** to the following questions and *submit as a single PDF on Gradescope* by the due date and time listed above. *Assign pages accordingly when you submit. It is recommended that you type as much as you can to ensure that your solution is legible. Illegible solutions will not receive credit.*

Note on Academic Dishonesty. Although you are allowed to discuss these problems with other people, your work must be entirely your own. It is considered academic dishonesty to read anyone else's solution to any of these problems or to share your solution with another student or to look for solutions to these questions online. See the syllabus for more information on academic dishonesty. If it is determined that you have violated any of these guidelines, you will be reported to the Dean of Students and the recommended sanction will be a failing grade in the course.

Grading. Some of these questions may be graded for accuracy and some for completion. Each question is worth 10 points.

Question 1.

Prove or disprove that if you have an 8-gallon jug of water and two empty jugs with capacities of 5 gallons and 3 gallons, respectively, then you can measure 4 gallons by successively pouring some of or all of the water in a jug into another jug.

Question 2.

Verify the $3x + 1$ conjecture for these integers.

a) 6 b) 7 c) 17 d) 21

Question 3.

Prove that you can use dominoes to tile a rectangular checkerboard with an even number of squares.

Question 4.

Show that by removing two white squares and two black squares from an 8×8 checkerboard (assume only black and white colors are used) you can make it impossible to tile the remaining squares using dominoes.

Question 5.

Find these terms of the sequence $\{a_n\}$, where $a_n = 2\cdot(-3)^n + 5^n$.

a) a_0 b) a_1 c) a_4 d) a_5

Question 6.

What is the term a_8 of the sequence $\{a_n\}$ if a_n equals

a) 2^{n-1}?

b) 7?

c) $1 + (-1)^n$?

d) $-(-2)^n$?

Question 7.

3. What are the terms a_0, a_1, a_2, and a_3 of the sequence $\{a_n\}$, where a_n equals

a) $2^n + 1$? b) $(n + 1)^{n+1}$?

c) $\lfloor n/2 \rfloor$? d) $\lfloor n/2 \rfloor + \lceil n/2 \rceil$?

Question 8.

List the first 10 terms of each of these sequences.

a) the sequence that begins with 2 and in which each successive term is 3 more than the preceding term

b) the sequence whose first term is 2, second term is 4, and each succeeding term is the sum of the two preceding terms

Question 9.

Find the first five terms of the sequence defined by each of these recurrence relations and initial conditions.

a) $a_n = 6a_{n-1}$, $a_0 = 2$

b) $a_n = a^2_{n-1}$, $a_1 = 2$

c) $a_n = a_{n-1} + 3a_{n-2}$, $a_0 = 1$, $a_1 = 2$

d) $a_n = na_{n-1} + n^2a_{n-2}$, $a_0 = 1$, $a_1 = 1$

e) $a_n = a_{n-1} + a_{n-3}$, $a_0 = 1$, $a_1 = 2$, $a_2 = 0$

Question 10.

Let $a_n = 2^n + 5\cdot 3^n$ for $n = 0, 1, 2, \ldots$

a) Find a_0, a_1, a_2, a_3, and a_4.

b) Show that $a_2 = 5a_1 - 6a_0$, $a_3 = 5a_2 - 6a_1$, and $a_4 = 5a_3 - 6a_2$.

c) Show that $a_n = 5a_{n-1} - 6a_{n-2}$ for all integers n with $n \geq 2$.

Question 11.

Show that the sequence $\{a_n\}$ is a solution of the recurrence relation $a_n = -3a_{n-1} + 4a_{n-2}$ if

a) $a_n = 0$. b) $a_n = 1$.

c) $a_n = (-4)^n$. d) $a_n = 2(-4)^n + 3$.

Question 12.

Is the sequence $\{a_n\}$ a solution of the recurrence relation $a_n = 8a_{n-1} - 16a_{n-2}$ if

a) $a_n = 0$? b) $a_n = 1$?

c) $a_n = 2^n$? d) $a_n = 4^n$?

Question 13.

For each of these sequences find a recurrence relation satisfied by this sequence. (The answers are not unique because there are infinitely many different recurrence relations satisfied by any sequence.)

a) $a_n = 3$ b) $a_n = 2n$

c) $a_n = 2n + 3$ d) $a_n = 5^n$

Question 14.

Show that the sequence $\{a_n\}$ is a solution of the recurrence relation $a_n = a_{n-1} + 2a_{n-2} + 2n - 9$ if

a) $a_n = -n + 2$.

b) $a_n = 5(-1)^n - n + 2$.

c) $a_n = 3(-1)^n + 2^n - n + 2$.

d) $a_n = 7 \cdot 2^n - n + 2$.

Question 15.

Find the solution to each of these recurrence relations with the given initial conditions. Use an iterative approach such as that used in class.

a) $a_n = -a_{n-1}$, $a_0 = 5$

b) $a_n = a_{n-1} + 3$, $a_0 = 1$

c) $a_n = a_{n-1} - n$, $a_0 = 4$

d) $a_n = 2a_{n-1} - 3$, $a_0 = -1$